

Rocks, Soil & Fossils

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Soil forms in **layers**: **topsoil** on top (dark, rich, with roots), **subsoil** below it, and **rock** at the bottom. Soil is a natural resource we need to grow food. **Fossils** are found in **sedimentary rock** layers. Fossils give us clues about living things from long ago.

Round 1 — Retrieve It

1. What is the top layer of soil called?

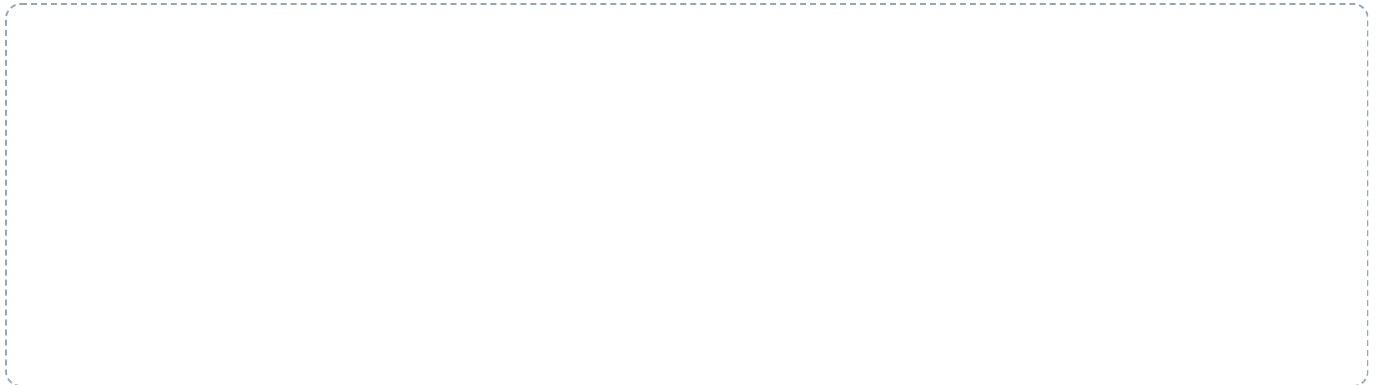
2. Name the three main layers of soil, from top to bottom.

3. In what kind of rock are fossils usually found?

4. What can fossils tell us?

Round 2 — Sketch It

Draw the layers of soil from top to bottom. Label topsoil, subsoil, and rock. Add a plant with roots reaching into the topsoil.



Rocks, Soil & Fossils — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. A farmer wants healthy crops. Which soil layer matters most for growing plants, and why?

2. A scientist finds the shape of a leaf pressed into a rock. What is this, and what does it show?

3. Why is soil called a natural resource? Give one way people depend on it.

Round 4 — Reason from Evidence

Scientists dig up a fish fossil high on a dry mountain. What might this evidence tell them about what that place was like long ago? Explain.

Rocks, Soil & Fossils — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain how soil forms in layers. Describe how the topsoil is different from the rock layer below.

C**Connect**

Connect fossils to sedimentary rock. Why do you think fossils are found in these layered rocks and not in most other rocks?

E**Extend**

Soil is a natural resource that takes a very long time to form. Suggest one way people could protect the soil in a garden or farm.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Rocks, Soil & Fossils

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Topsoil.
2. Topsoil, subsoil, and rock (bedrock).
3. Sedimentary rock.
4. They give clues about living things (plants and animals) from long ago and what the environment was like.

Sketch It — look for

Three layers drawn in order: topsoil on top, subsoil in the middle, rock at the bottom, with a plant whose roots reach into the dark topsoil.

Apply It

1. Topsoil; it is rich in nutrients and holds water and roots, so plants grow best there.
2. A fossil; it shows that a plant like that lived there long ago.
3. Soil is a natural resource because it comes from the Earth and people use it to grow food and plants.

Reason from Evidence — look for

A fish fossil suggests that long ago the area may have been covered by water (a lake, river, or ocean), even though it is dry land now.

ACE — Extend look-for

A reasonable way to protect soil, such as planting cover plants, not overusing it, avoiding erosion, or adding compost to keep it rich.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.